

Turn a finished analysis into a submission-ready publication — automatically, and in every format the journal wants.

We built this to stop hand-writing NIEHS toxicology reports. Along the way it turned into something more general: a system that takes structured scientific data and emits a complete, correctly-formatted paper. This is a discussion draft on whether that's worth (a) leaning into internally and (b) offering as a service.

The one-liner

Scientists spend enormous effort turning correct results into a correctly *formatted* document. We have automated that last mile for one demanding report type — data in, a publication-grade document out, in Word, PDF, LaTeX/Overleaf, and the XML a government publisher requires. The pitch: the last mile is nearly the same for *any* data-driven paper, and almost nobody has automated it well.

1 · The problem we solved

The U.S. National Toxicology Program runs a fast, data-rich chemical-safety study. For every chemical, a scientist must turn the results into a **biological potency report**: a long, rigidly conventional document — dozens of statistical tables that must follow an exact house style, narrative sections, a genomic interpretation tying gene-level signals to the classic toxicology findings, and a reference layout (*NIEHS Report 10*) every report is measured against.

Writing one by hand is slow, repetitive, and error-prone. The *science* is finished when the analysis is done; what remains is weeks of formatting, table construction, cross-checking numbers, and fighting the document until it matches the template. That "last mile" is pure cost — and it is exactly the kind of work that punishes humans and rewards automation.

2 · What exists today (not a mock-up)

This is a working system, in daily development, that has produced real reports. Concretely, it already does the following:

Ingests the raw study

Loads the analytical outputs, cross-validates the files, and merges them into one canonical dataset — catching conflicts a human would miss.

Computes the report content

Runs the required statistics and dose-response modeling, builds every table to the house style, and generates the narrative — grounded in the data, not free-text guesses.

Emits four formats from one source

Word (`.docx`), print-ready PDF, LaTeX/Overleaf, and the Bookshelf **JATS/BITS XML** a publisher ingests — all from a single structural definition, so they never drift apart.

Matches a gold-standard layout

Output is measured page-by-page against the reference report, down to table numbering, footnotes, and section ordering.

Import

raw study files,
validated & merged

Process

stats, modeling, tables,
narrative

Assemble

one document
structure, many
sections

Publish

Word · PDF · LaTeX ·
publisher XML

Two architectural choices matter for the pitch, because they are what make the thing *generalize* rather than being a one-off script:

- **One structure drives every format.** The document is defined once, as a structure; each output format is a projection of that one definition. Add a section once, get it in all four formats.
- **Content is bound, not baked.** Each section of the document declares *what data it needs* and *what shape* that data takes; the system fills it in. The document is a set of slots with a contract, not a hard-coded template.

3 · The generalizing insight

Here is the step from "a tool for one report" to "a platform." The hard, report-specific part is the toxicology math. But the **last mile — structured results → formatted, multi-format, submission-ready document — is almost identical for every data-driven paper**. A clinical study, a materials-science characterization, an economics working paper, a recurring internal technical report: all of them are "run the numbers, then spend weeks making the document."

Because our document is a set of *slots with a declared data shape*, the path to a new kind of paper is: describe the new document's structure and the shape of data each section expects, then point it at a data source that produces that shape. The formatting, the multi-format output, the layout fidelity — all of that is already built and is *shared*.

*The vision, stated plainly: **start from any data-processing pipeline and generate a paper.***

The disciplined version of that vision. We are deliberately *not* promising "we'll integrate any pipeline anyone brings us" — that's an endless custom-integration treadmill and a bad business. Instead: the **document declares the shape it needs, and data conforms to that shape**. The upstream pipeline stays a black box we don't maintain. That distinction is the difference between a product and a consultancy. One-off, bespoke integrations are explicitly not worth it; a clean shape contract that many pipelines can target is.

4 · The internal case (streamline our own pipeline)

Independent of any commercial question, this pays for itself internally:

Where the time goes today	What the platform changes
Hand-building statistical tables to an exact house style	Generated to the reference layout, every time, with correct numbering
Re-formatting the same content for Word, PDF, and the publisher's XML	One structure, all formats emitted together — no re-keying, no drift
Manually cross-checking numbers between analysis and document	Numbers flow from the data; the document can't disagree with the source
A specialist bottleneck who "knows how to make the document"	Institutional knowledge encoded once; the next author inherits it

Succession value. The formatting expertise for these reports currently lives in a small number of heads. Encoding it in the platform is insurance: it survives staff turnover and onboards the next scientist without re-learning a decade of conventions.

The near-term internal win is a self-service power-user capability we're already building the foundation for: a scientist can ask questions of a study's data directly, and — further out — define a new report section by pointing it at a query, without waiting on an engineer. That shortens the loop from "we need a new kind of table" to a same-day change.

5 • The service case (offer it externally)

The same capability, pointed outward, is a service. Three plausible shapes, from lightest to heaviest lift:

1 · Formatting-as-a-service closest

Customer brings structured results in an agreed shape; we return a submission-ready document in every format the target venue needs. We own the last mile; they own their science.

2 · Template packs adjacent

Pre-built document definitions for common report types (a study type, a journal family, a regulatory filing). The customer's data drops into a proven structure.

3 · Hosted platform largest

A self-serve product: upload data, choose or define a document, publish. The internal power-user tools become the customer-facing product.

All three ride on the same engine. The service question is really about *packaging and support*, not about building something new — which is the healthiest possible position to pitch from.

6 · Who would buy this

- **Government & regulatory science programs** — anyone who must publish to a fixed house style or a mandated XML format (Bookshelf, regulatory submissions). This is exactly the pain we already solve, and the format requirements are non-negotiable, which is precisely where automation wins.
- **Contract research organizations (CROs)** — they produce the same report shape over and over for many clients. Per-report margin improvement scales directly with volume.
- **Core facilities & large labs** — recurring internal technical reports with a standard structure and a standard analysis.
- **Scientific publishers / platforms** — a partner that can take authors from data to a spec-conformant submission reduces their editorial and correction burden.

The wedge is format compliance. Our sharpest, most defensible edge is producing the *exact* structured output a publisher or regulator demands — the part authors hate most and general-purpose tools handle worst. Lead with that, not with "we write papers."

7 · Why it's defensible

- **Multi-format from one source is hard to fake.** Keeping Word, PDF, LaTeX, and publisher-XML byte-faithful to one definition is real engineering most teams underestimate; we've already paid that cost.
- **Layout fidelity is a specification, not a vibe.** We measure output against a gold-standard reference. "Close enough" is what free tools do; "passes the publisher's validator" is what customers pay for.
- **Data-grounded content.** Tables and narrative derive from the data, so the document can't silently disagree with the analysis — a correctness property that matters enormously in regulated science and that pure LLM "write my paper" tools cannot guarantee.
- **Domain credibility.** Born inside a demanding real-world reporting program, validated against its gold standard — not a generic demo.

8 · What's real vs. what's roadmap

So the conversation stays grounded — a clear line between shipped and proposed:

Capability	Status
End-to-end report generation for our study type (import → four formats)	Working today
Multi-format output from one document structure	Working today
Layout measured against the gold-standard reference	Working today
Multiple report versions from one processed dataset	Working today
Self-service data query for power users	Designed, building
Define a new section by pointing it at a query ("rendering domains")	Designed
Ingesting an external pipeline's output via a shape contract	Proposed
New non-toxicology paper types	Vision

Honest risks to name in the room. (1) Generalizing beyond our own report type is real work, not a config change — the first new paper type is the expensive one. (2) A service means support, SLAs, and customer data handling we don't do today. (3) The market wedge is format *compliance*; if we drift into "AI writes your paper," we're in a crowded, less-defensible fight. The strategy is to stay narrow and deep where we're strong.

9 • The ask

Nothing here needs a big bet to explore. What would help:

1. **A read on the market.** Is "submission-ready formatting + mandated-format compliance, from structured data" a pain people pay to remove? Which of the buyer segments in §6 is worth a first conversation?
2. **One reference customer to shape against.** A single external (or internal-adjacent) report type to prove the generalization on — the fastest way to learn whether the shape-contract approach holds outside our own walls.
3. **Positioning help.** If we pursue this, the framing ("the last mile of publication," "format compliance as a service") matters as much as the technology. That's your expertise, not ours.

The engine exists and works. The question this document puts on the table is strategic, not technical: **how far do we push it, and outward to whom?**

Internal discussion draft. Capability claims reflect the working system as of August 2026; items marked *Designed*, *Proposed*, or *Vision* are roadmap, not shipped. Technical backing: ADR-0016 (query substrate & rendering domains), the data-integration and filtering-system docs, and the four live render surfaces.